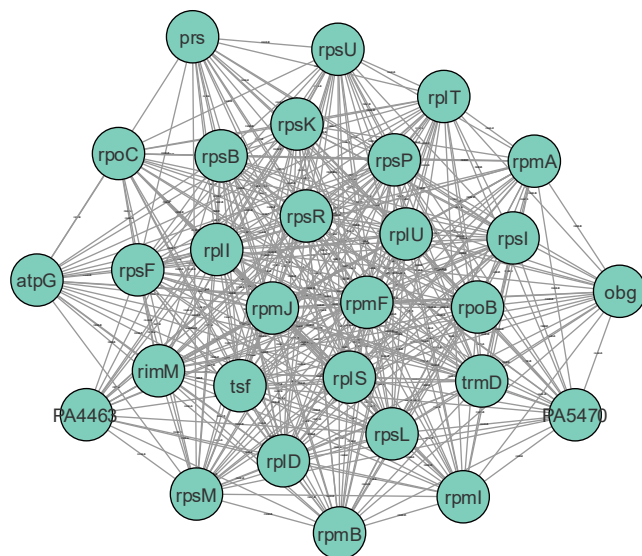


### Functional modules identified by MCODE analysis.

## A Cluster 1 - Ribosome Biogenesis and Translation Machinery

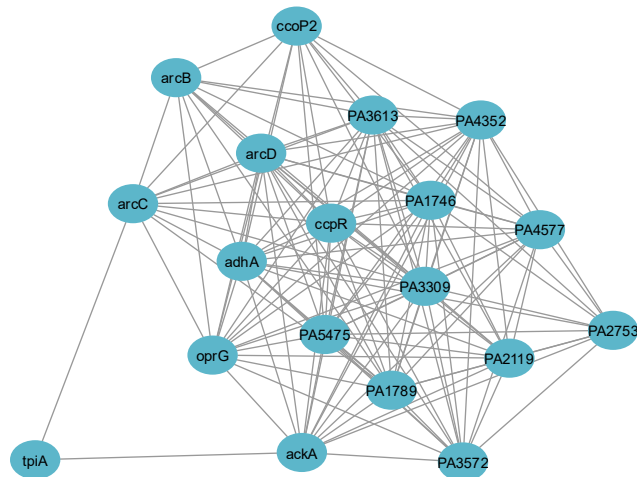


**Score (Density\*#Nodes): 26.786**

Nodes: 29 Edges: 375

## Ribosomal proteins and translation-associated factors involved in protein synthesis and cellular adaptation

## B Cluster 2 – Metabolic Adaptation and Anaerobic Stress Response

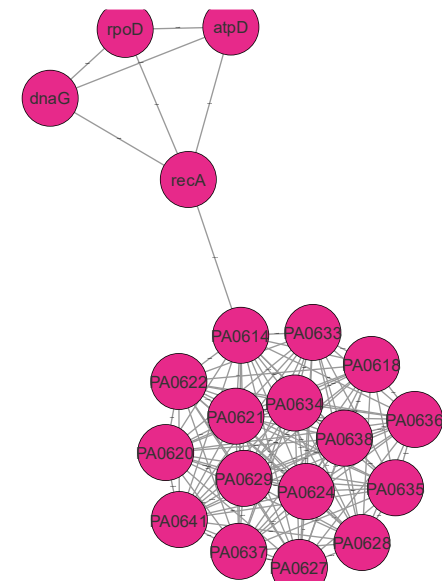


**Score (Density\*#Nodes): 14.889**

**Nodes: 19      Edges: 134**

**Metabolic and anaerobic stress-response proteins involved in energy homeostasis and arginine catabolism.**

### **C Cluster 3 – Tailocin (R-pyocin)-Associated Module**



**Score (Density\*#Nodes): 13.368**

Nodes: 20 Edges: 127

## Tailocin-associated and phage tail-like contractile proteins involved in interbacterial competition.

**MCODE:** Molecular Complex Detection    • **recA:** DNA repair and SOS response    • **dnaG:** DNA replication initiation  
**rpoB, rpoC:** RNA polymerase and transcription machinery    • **rpl/rps/rpm:** Ribosomal proteins and translation machinery  
**arcB, arcC, arcD:** Arginine deiminase pathway    • **ccoP2:** Microaerobic respiration    • **tpiA, ackA, adhA:** Central carbon metabolism